

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                       |
|----------------------|-----------------------|
| United States Patent | 12408878              |
| Kind Code            | B2                    |
| Date of Patent       | September 09, 2025    |
| Inventor(s)          | Huang; Chenhui et al. |

---

### Risk estimation apparatus, risk estimation system, risk estimation method, and non-transitory computer readable recording medium

---

#### Abstract

A risk estimation apparatus includes a data acquisition unit that acquires measured data of foot pressures of the left foot and the right foot obtained by sensors that are provided in shoes and measure the foot pressures, a stance phase identification unit that identifies a starting timing and an ending timing of a stance phase of each of the left foot and the right foot from the measured data of the foot pressures, and a risk estimation unit that estimates a risk of abnormality of a lower limb on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

---

**Inventors:** Huang; Chenhui (Tokyo, JP), Fukushi; Kenichiro (Tokyo, JP), Sekiguchi; Yusuke (Sendai, JP), Yaguchi; Haruki (Sendai, JP), Honda; Keita (Sendai, JP), Izumi; Shinichi (Sendai, JP), Owaki; Dai (Sendai, JP)

**Applicant:** NEC Corporation (Tokyo, JP)

**Family ID:** 79166417

**Assignee:** NEC CORPORATION (Tokyo, JP)

**Appl. No.:** 18/413378

**Filed:** January 16, 2024

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240148336 A1          | May. 09, 2024           |

#### Foreign Application Priority Data

|    |             |               |
|----|-------------|---------------|
| JP | 2020-115941 | Jul. 03, 2020 |
|----|-------------|---------------|

Related U.S. Application Data

continuation parent-doc US 17362498 20210629 US 11925483 child-doc US 18413378

Publication Classification

Int. Cl.: A61B5/00 (20060101); A61B5/103 (20060101); G16H40/67 (20180101); G16H50/30 (20180101)

U.S. Cl.:

CPC A61B5/7275 (20130101); A61B5/1038 (20130101); A61B5/6807 (20130101); G16H40/67 (20180101); G16H50/30 (20180101); A61B2562/0247 (20130101); A61B2562/04 (20130101)

Field of Classification Search

CPC: A61B (5/7275); A61B (5/1038); A61B (5/6807); A61B (2562/0247); A61B (2562/04); A61B (5/112); G16H (40/67); G16H (50/30)

References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name         | U.S. Cl. | CPC         |
|--------------|-------------|-----------------------|----------|-------------|
| 10555689     | 12/2019     | Marquez et al.        | N/A      | N/A         |
| 11925483     | 12/2023     | Huang                 | N/A      | G16H 40/67  |
| 2009/0240171 | 12/2008     | Morris Bamberg et al. | N/A      | N/A         |
| 2014/0195023 | 12/2013     | Statham et al.        | N/A      | N/A         |
| 2019/0174862 | 12/2018     | Rakshit               | N/A      | A43B 3/38   |
| 2020/0151594 | 12/2019     | Schwartz              | N/A      | A61B 5/7267 |

FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 2009-106375 | 12/2008          | JP      | N/A |
| 2016-106973 | 12/2015          | JP      | N/A |
| 2018/164157 | 12/2017          | WO      | N/A |

OTHER PUBLICATIONS

Wafai L, Zayegh A, Woulfe J, Aziz SM, Begg R. Identification of Foot Pathologies Based on Plantar Pressure Asymmetry. Sensors (Basel). Aug. 18, 2015;15(8):20392-408. doi: 10.3390/s150820392. PMID: 26295239; PMCID: PMC4570427., (Year: 2015). cited by examiner JP Office Communication for JP Application No. 2020-115941, mailed on May 21, 2024 with English Translation. cited by applicant JP Office Action for JP Application No. 2020-115941, mailed on Dec. 5, 2023 with English Translation. cited by applicant Wang C, Kim Y, Shin H, Min SD. Preliminary Clinical Application of Textile Insole Sensor for Herniparetic Gait Pattern Analysis. Sensors (Basel). Sep. 12, 2019;19(18):3950. doi:

*Primary Examiner:* Robertson; Jennifer

*Assistant Examiner:* Jang; Elina Sohyun

---

## **Background/Summary**

**CROSS-REFERENCE TO RELATED APPLICATION** (1) This application is a Continuation of U.S. application Ser. No. 17/362,498, filed Jun. 29, 2021, which is based upon and claims the benefit of priority from Japanese Patent Application No. 2020-115941, filed Jul. 3, 2020, the disclosure of which is incorporated herein in its entirety by reference.

### **TECHNICAL FIELD**

(1) The present invention relates to a risk estimation apparatus, a risk estimation system, a risk estimation method, and a non-transitory computer readable recording medium.

### **BACKGROUND ART**

(2) A technology of providing sensors in shoes and performing evaluation related to the feet has been proposed.

(3) For example, in the system disclosed in PCT International Publication No. WO 2018/164157 (hereinafter referred to as “Patent Document 1”), data obtained using sensors provided in insoles of shoes is stored for each user. In this system, data similar to measured data is selected from stored data, and a likelihood of occurrence of abnormality in the feet, such as the presence or absence of formation of calluses, when a user continuously wears shoes is determined using information that is indicated for the selected data and is related to the feet and the shoes, such as that regarding formation of calluses.

### **SUMMARY**

(4) It is preferable to be able to estimate a risk related to the feet without being limited to a relationship between the feet and shoes.

(5) An example object of the present invention is to provide a risk estimation apparatus, a risk estimation system, a risk estimation method, and a non-transitory computer-readable recording medium capable of resolving the above-described problems.

(6) A first example aspect of the present invention is a risk estimation apparatus including: at least one memory configured to store instructions; and at least one processor configured to execute the instructions to: acquire measured data of foot pressures of a left foot and a right foot obtained by sensors that are provided in shoes and measure the foot pressures; identify a starting timing and an ending timing of a stance phase of each of the left foot and the right foot from the measured data of the foot pressures; and estimate a risk of abnormality of a lower limb on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

(7) A second example aspect of the present invention is a risk estimation system including: sensors that are provided in shoes and measure foot pressures of a left foot and a right foot; at least one memory configured to store instructions; and at least one processor configured to execute the instructions to: identify a starting timing and an ending timing of a stance phase of each of the left foot and the right foot from measured data of the foot pressures; and estimate a risk of abnormality of a lower limb on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

(8) A third example aspect of the present invention is a risk estimation method including: acquiring, by a risk estimation apparatus, measured data of foot pressures of a left foot and a right foot

obtained by sensors that are provided in shoes and measure the foot pressures; identifying, by the risk estimation apparatus, a starting timing and an ending timing of a stance phase of each of the left foot and the right foot from the measured data of the foot pressures; and estimating, by the risk estimation apparatus, a risk of abnormality of a lower limb on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

(9) According to the present invention, it is possible to estimate a risk related to the feet other than a relationship between the feet and shoes.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram illustrating an example of configurations of apparatuses in a risk estimation system according to an example embodiment.

(2) FIG. 2 is a schematic block diagram illustrating an example of a functional configuration of the risk estimation system according to the example embodiment.

(3) FIG. 3 is a first diagram illustrating an example of disposition of sensors of a sensor system according to the example embodiment.

(4) FIG. 4 is a second diagram illustrating an example of disposition of sensors of the sensor system according to the example embodiment.

(5) FIG. 5 is a diagram illustrating an example of a stance phase and a swing phase in one walking cycle.

(6) FIG. 6 is a diagram illustrating an example of measured values of foot pressures obtained by the sensor system according to the example embodiment.

(7) FIG. 7 is a diagram illustrating an example of risk level information stored in a storage unit according to the example embodiment.

(8) FIG. 8 is a flowchart illustrating an example of a procedure of processing in which a risk estimation apparatus according to the example embodiment estimates a risk of abnormality of a lower limb (hereinafter referred to as “a lower limb abnormality risk”).

(9) FIG. 9 is a flowchart illustrating an example of a procedure of processing in which the risk estimation apparatus according to the example embodiment synchronizes data of a left side sensor system with data of a right side sensor system.

(10) FIG. 10 is a flowchart illustrating an example of a procedure of processing in which the risk estimation apparatus according to the example embodiment determines the stance phases for data of the left side sensor system and data of the right side sensor system.

(11) FIG. 11 is a diagram illustrating another example of a configuration of the risk estimation system according to the example embodiment.

(12) FIG. 12 is a diagram illustrating an example of a configuration of the risk estimation apparatus according to the example embodiment.

(13) FIG. 13 is a diagram illustrating an example of a configuration of the risk estimation system according to the example embodiment.

(14) FIG. 14 is a flowchart illustrating an example of a procedure of processing in a risk estimation method of the risk estimation apparatus according to the example embodiment.

(15) FIG. 15 is a schematic block diagram illustrating a configuration of a computer according to at least one of the example embodiments.

### EXAMPLE EMBODIMENT

(16) Hereinafter, example embodiments of the present invention will be described but the following example embodiments do not limit the invention according to the claims. Moreover, not all the combinations of features described in the example embodiments are essential for the solution of the invention.

(17) FIG. 1 is a schematic configuration diagram illustrating an example of a configurations of apparatuses in a risk estimation system according to the example embodiment. In the configuration illustrated in FIG. 1, a risk estimation system **1** includes a left side sensor system **101**, a right side sensor system **102**, and a risk estimation apparatus **200**.

(18) The risk estimation system **1** measures foot pressures of a subject person of risk estimation and estimates a lower limb abnormality risk the subject person of risk estimation on the basis of measured data. Here, foot pressures indicate pressures applied to the soles of the feet. Foot pressures are generated due to the body weight supported by the feet. Here, a lower limb abnormality risk indicates the degree of likelihood of occurrence of abnormality in a lower limb. Moreover, a subject person of risk estimation is also simply referred to as a subject person.

(19) Specifically, the risk estimation system **1** measures foot pressures of the left foot and the right foot when the subject person walks and estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot. Here, asymmetry between the foot pressures of the left foot and right foot indicates that the foot pressures during the same phase of walking motion differ between the left foot and the right foot.

(20) Here, if loads applied to the feet are small, bone densities of the feet decrease. If the bone densities of the feet decrease, a lower limb abnormality risk increases. Thus, if loads applied to the feet are small, a lower limb abnormality risk increases.

(21) Hence, the risk estimation system **1** estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot. It is thought that the bone density of the foot on a side of a small foot pressure decreases and a lower limb abnormality risk increases due to asymmetry between the foot pressures of the left foot and right foot.

(22) The left side sensor system **101** is provided in a shoe on the left foot side and measures the foot pressure of the left foot. The left side sensor system **101** transmits the measured data of the foot pressure of the left foot to the risk estimation apparatus **200**.

(23) The right side sensor system **102** is provided in a shoe on the right foot side and measures the foot pressure of the right foot. The right side sensor system **102** transmits the measured data of the foot pressure of the right foot to the risk estimation apparatus **200**.

(24) The left side sensor system **101** and the right side sensor system **102** will be collectively referred to as sensor systems **100**. The sensor systems **100** correspond to an example of sensors that are provided in shoes and measure the foot pressures of the left foot and the right foot.

(25) Shoes provided with the sensor systems **100** will be referred to as shoes **810**. Of the shoes **810**, a shoe for the left foot will be referred to as a left foot shoe **811**. The left foot shoe **811** is provided with the left side sensor system **101**. Of the shoes **810**, a shoe for the right foot will be referred to as a right foot shoe **812**. The right foot shoe **812** is provided with the right side sensor system **102**.

(26) The risk estimation apparatus **200** estimates a lower limb abnormality risk of the subject person on the basis of asymmetry between the foot pressures of the left foot and the right foot of the subject person. Specifically, the risk estimation apparatus **200** receives the measured data of the foot pressures of the left foot and the right foot from the left side sensor system **101** and the right side sensor system **102**. Then, the risk estimation apparatus **200** calculates an index value of asymmetry between the foot pressures of the left foot and the right foot on the basis of the obtained measured data. The risk estimation apparatus **200** calculates a lower limb abnormality risk on the basis of the index value of asymmetry between the foot pressures of the left foot and the right foot.

(27) The risk estimation apparatus **200** is configured using a portable computer such as a smartphone or a tablet personal computer (tablet PC). In this manner, the risk estimation apparatus **200** is configured using a portable computer and the subject person carries the risk estimation apparatus **200**, and thus the risk estimation apparatus **200** is continuously positioned near the sensor systems **100**. Accordingly, it is relatively easy for the risk estimation apparatus **200** and the sensor systems **100** to communicate with each other, such as in a case in which the risk estimation apparatus **200** and the sensor systems **100** communicate with each other through, for example,

short-range wireless communication.

(28) FIG. 2 is a schematic block diagram illustrating an example of a functional configuration of the risk estimation system 1. FIG. 2 illustrates a functional configuration of the left side sensor system 101, the right side sensor system 102, and the risk estimation apparatus 200 illustrated in FIG. 1. The left side sensor system 101 includes a left toe side sensor 111, a left heel side sensor 121, and a left side communication apparatus 131. The right side sensor system 102 includes a right toe side sensor 112, a right heel side sensor 122, and a right side communication apparatus 132. The risk estimation apparatus 200 includes a communication unit 210, a display unit 220, an operation input unit 230, a storage unit 280, and a control unit 290. The control unit 290 includes a data acquisition unit 291, a synchronization unit 292, a stance phase identification unit 293, and a risk estimation unit 294.

(29) The left toe side sensor 111 and the right toe side sensor 112 will be collectively referred to as toe side sensors 110. The left heel side sensor 121 and the right heel side sensor 122 will be collectively referred to as heel side sensors 120. The left side communication apparatus 131 and the right side communication apparatus 132 will be collectively referred to as communication apparatuses 130.

(30) The left toe side sensor 111 is provided on the toe side of the left foot shoe 811 and measures the foot pressure on the toe side of the left foot. In particular, the left toe side sensor 111 measures the foot pressure when the toes of the left foot take off the ground.

(31) The left heel side sensor 121 is provided on the heel side of the left foot shoe 811 and measures the foot pressure on the heel side of the left foot. In particular, the left heel side sensor 121 measures the foot pressure when the heel of the left foot contacts the ground. Here, a state in which a foot contacts the ground may be a state in which a shoe contacts the ground. That is, a state in which a shoe contacts the ground may be referred to as a state in which a foot contacts the ground. Moreover, a state in which a foot takes off the ground may be a state in which a shoe takes off the ground. That is, a state in which a shoe takes off the ground may be referred to as a state in which a foot takes off the ground.

(32) The left side communication apparatus 131 transmits the measured data of the left toe side sensor 111 and the left heel side sensor 121 to the risk estimation apparatus 200.

(33) The right toe side sensor 112 is provided on the toe side of the right foot shoe 812 and measures the foot pressure on the toe side of the right foot. In particular, the right toe side sensor 112 measures the foot pressure when the toes of the right foot take off the ground.

(34) The right heel side sensor 122 is provided on the heel side of the right foot shoe 812 and measures the foot pressure on the heel side of the right foot. In particular, the right heel side sensor 122 measures the foot pressure when the heel of the right foot contacts the ground.

(35) The right side communication apparatus 132 transmits the measured data of the right toe side sensor 112 and the right heel side sensor 122 to the risk estimation apparatus 200.

(36) FIG. 3 is a first diagram illustrating an example of disposition of sensors of the sensor system 100. FIG. 3 illustrates an example of disposition of sensors of the sensor system 100 when the shoe 810 is viewed in a lateral direction.

(37) In the example of FIG. 3, the toe side sensor 110 is provided on the toe side, and the heel side sensor 120 is provided on the heel side. In particular, in the example of FIG. 3, a portion of the shoe 810 that contacts the ground is divided into a toe portion on the toe side (the front side of the shoe 810) and a heel portion on the heel side (the rear side of the shoe 810). The toe side sensor 110 is provided in the toe portion of the shoe 810, and the heel side sensor 120 is provided in the heel portion of the shoe 810.

(38) The communication apparatus 130 is provided between the toe portion and the heel portion of the shoe 810.

(39) It should be noted that FIG. 3 illustrates an example of a case in which the toe side sensor 110 and the heel side sensor 120 are provided in the vicinity of a bottom surface (a surface coming into

contact with the ground) of the shoe, but disposition of the toe side sensor **110** and the heel side sensor **120** is not limited thereto. For example, holes for respectively storing the toe side sensor **110** and the heel side sensor **120** may be provided on an upper surface (e.g., immediately below an insole (a shoe insert)) of a shoe sole, and the toe side sensor **110** and the heel side sensor **120** may be respectively fitted into the holes. Alternatively, the toe side sensor **110** and the heel side sensor **120** may be provided in the insole.

(40) Moreover, regarding disposition of the communication apparatus **130**, the position of the communication apparatus **130** need only be at a position where measured data can be acquired from the toe side sensor **110** and the heel side sensor **120** and the acquired measured data can be transmitted to the risk estimation apparatus **200**.

(41) FIG. 4 is a second diagram illustrating an example of disposition of sensors of the sensor system **100**. FIG. 4 illustrates an example of disposition of sensors of the left side sensor system **101** when the left foot shoe **811** is viewed from the shoe sole side. That is, FIG. 4 illustrates an example of disposition of sensors of the left side sensor system **101** when the shoe sole of the left foot shoe **811** is viewed in a manner of being looked up at from the ground side during walking.

(42) In the example of FIG. 4, the left foot shoe **811** is divided into the toe side and the heel side with a line **L11** as a boundary. The left toe side sensor **111** is provided on the toe side of the left foot shoe **811**. The left heel side sensor **121** is provided on the heel side of the left foot shoe **811**.

(43) The right side sensor system **102** may be provided in a bilaterally symmetrical manner with the example of FIG. 4. The right foot shoe **812** is also divided into the toe side and the heel side, the right toe side sensor **112** may be provided on the toe side of the right foot shoe **812**, and the right heel side sensor **122** may be provided on the heel side of the right foot shoe **812**.

(44) Regarding setting of the toe side and the heel side of the shoe **810**, they need only be set such that the toe side sensor **110** is disposed at a position where the foot pressure can be measured when the toes take off the ground and the heel side sensor **120** is disposed at a position where the foot pressure can be measured when the heel contacts the ground.

(45) For example, the toe side and the heel side of the shoe **810** may be set on the basis of the structure of the shoe **810** as in the example of FIG. 3.

(46) Alternatively, the toe side and the heel side of the shoe **810** may be set on the basis of the structure of the foot. For example, with respect to the position of the arch of the sole of the foot when the subject person puts on the shoe **810**, the toe side from the arch of the sole of the foot may be set as the toe side of the shoe **810**, and the heel side from the arch of the sole of the foot may be set as the heel side of the shoe **810**. Alternatively, with respect to a position of the joint portion between the metatarsal bone and the tarsal bone when the subject person puts on the shoe **810**, the toe side from the joint portion between the metatarsal bone and the tarsal bone may be set as the toe side of the shoe **810**, and the heel side from the joint portion between the metatarsal bone and the tarsal bone may be set as the heel side of the shoe **810**.

(47) The number of sensors included in the sensor system **100** is not limited to two. Depending on a method by which the risk estimation apparatus **200** calculates the index value of asymmetry between the foot pressures of the left foot and the right foot, the sensor system **100** may include only one of the toe side sensor **110** and the heel side sensor **120**. That is, a sensor may be provided on only one of the toe side and the heel side of the shoe **810**.

(48) Alternatively, the sensor system **100** may include three or more sensors. Alternatively, the sensor system **100** may include one sensor capable of measuring the foot pressures on both the toe side and the heel side.

(49) When the sensor system **100** includes the toe side sensor **110** and the heel side sensor **120**, the communication apparatus **130** may transmit each of the measured data of the toe side sensor **110** and the measured data of the heel side sensor **120** to the risk estimation apparatus **200**.

Alternatively, the communication apparatus **130** may transmit data of foot pressures obtained by summing the foot pressure measured by the toe side sensor **110** and the foot pressure measured by

the heel side sensor **120** to the risk estimation apparatus **200**.

(50) The communication unit **210** of the risk estimation apparatus **200** communicates with other apparatuses. In particular, the communication unit **210** communicates with the left side communication apparatus **131** and acquires the measured data of the foot pressure of the left foot. Moreover, the communication unit **210** communicates with the right side communication apparatus **132** and acquires the measured data of the foot pressure of the right foot.

(51) The communication scheme of the risk estimation apparatus **200** is not limited to a particular scheme. For example, the risk estimation apparatus **200** may communicate with each of the left side communication apparatus **131** and the right side communication apparatus **132** by means of a communication scheme of short-range wireless communication, but it is not limited thereto.

(52) For example, the display unit **220** includes a display screen such as a liquid crystal panel or a light emitting diode (LED) panel and displays various images. For example, the display unit **220** displays estimation results of a lower limb abnormality risk of the subject person.

(53) However, a method by which the risk estimation apparatus **200** outputs data is not limited to the method in which the display unit **220** displays data. For example, the communication unit **210** may transmit estimation results of a lower limb abnormality risk of the subject person to other apparatuses such as a server apparatus.

(54) For example, the operation input unit **230** includes an input device such as a touch sensor that configures a touch panel provided in the display screen of the display unit **220** and receives an operation of a user. For example, the operation input unit **230** receives an operation of a user instructing estimation of a lower limb abnormality risk.

(55) The storage unit **280** stores various pieces of data. The storage unit **280** is configured using a storage device included in the risk estimation apparatus **200**.

(56) The control unit **290** performs various kinds of processing by controlling each unit of the risk estimation apparatus **200**. For example, the functions of the control unit **290** are executed by a central processing unit (CPU) included in the risk estimation apparatus **200** reading a program from the storage unit **280** and executing the program.

(57) The data acquisition unit **291** acquires the measured data of the foot pressure of the left foot obtained by the left side sensor system **101** and the measured data of the foot pressure of the right foot obtained by the right side sensor system **102**. Specifically, the data acquisition unit **291** extracts the measured data of the foot pressure of the left foot from a reception signal that is a signal received by the communication unit **210** from the left side sensor system **101**. Moreover, the data acquisition unit **291** extracts the measured data of the foot pressure of the right foot from a reception signal that is a signal received by the communication unit **210** from the right side sensor system **102**.

(58) The data acquisition unit **291** may acquire the measured data of the foot pressures on the heel side and the toe side of each of the left foot and the right foot. For example, the data acquisition unit **291** may acquire the measured data of the toe side sensor **110** and the measured data of the heel side sensor **120**. Alternatively, the data acquisition unit **291** may acquire data obtained by summing a measured value of the foot pressure by the toe side sensor **110** and a measured value of the foot pressure by the heel side sensor **120**.

(59) The synchronization unit **292** synchronizes the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot. Here, synchronizing the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot indicates associating the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot with respect to at least parts of measurement periods that temporally overlap each other. For example, synchronizing the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot may be associating data for one walking cycle of the left foot with data for one walking cycle of the right foot with respect to parts of measurement periods that overlap each other.

(60) The stance phase identification unit **293** identifies a starting timing and an ending timing of a stance phase of each of the left foot and the right foot from the measured data of the foot pressure of each of the left foot and the right foot.

(61) FIG. 5 is a diagram illustrating an example of a stance phase and a swing phase in one walking cycle. The horizontal axis in FIG. 5 indicates times. The horizontal axis in FIG. 5 indicates times in which one walking cycle is normalized to 100.

(62) In the example of FIG. 5, the right foot (the right foot portion) contacts the ground at a normalized time **0**, the right foot takes off the ground at a normalized time **60**, and the right foot contacts the ground again at a normalized time **100**. In the example of FIG. 5, when the foot contacts the ground, the foot contacts the ground from the heel. Moreover, when the foot takes off the ground, the toes take off the ground last. For this reason, the heel of the right foot contacts the ground at the normalized time **0**, and the toes of the right foot take off the ground at the normalized time **60**. The heel of the right foot contacts the ground again at the normalized time **100**.

(63) In this manner, the foot contacts the ground from the heel during ordinary walking, and the toes take off the ground last when the foot takes off the ground.

(64) A period from when the foot contacts the ground until the foot takes off the ground will be referred to as a stance phase. Moreover, a period from when the foot takes off the ground until the foot contacts the ground will be referred to as a swing phase. In the example of FIG. 5, the period from the normalized time **0** to the normalized time **60** corresponds to the stance phase of the right foot. The period from the normalized time **60** to the normalized time **100** corresponds to the swing phase of the right foot.

(65) Regarding the left foot (the left foot portion), the left foot takes off the ground at a normalized time **10**, and the left foot contacts the ground at a normalized time **50**. Each of the period to the normalized time **10** and the period from the normalized time **50** corresponds to the stance phase of the left foot. The period from the normalized time **10** to the normalized time **50** corresponds to the swing phase of the left foot. Regarding also the left foot, the foot contacts the ground from the heel, and the toes take off the ground last.

(66) In the example of FIG. 5, one walking cycle is a period from when the right foot contacts the ground until the right foot contacts the ground next. However, one walking cycle is not limited thereto and need only be a period from a certain phase in either the left or right foot to the next same phase in the same foot. For example, one walking cycle may be a period from when the left foot contacts the ground until the left foot contacts the ground again next. Alternatively, one walking cycle may be a period from when the right foot takes off the ground until the right foot takes off the ground again next.

(67) In the example of FIG. 5, in the period from the normalized time **0** to the normalized time **60**, which is the stance phase of the right foot, and the period from the normalized time **50**, which is the stance phase of the left foot, the periods from the normalized time **50** to the normalized time **60** temporally overlap each other. In this manner, parts of the stance phase of the right foot and the stance phase of the left foot overlap each other during walking.

(68) For example, the synchronization unit **292** associates pieces of the measured data in which at least parts of the measurement periods thereof overlap with each other; for example, associates the measured data of the foot pressure of the right foot during the period from the normalized time **0** to the normalized time **60**, which is the stance phase of the right foot, with the measured data of the foot pressure of the left foot during the period from the normalized time **50**, which is the stance phase of the left foot.

(69) Assuming a case in which a measurement time of the measured data of the foot pressure of the left foot significantly differs from a measurement time of the measured data of the foot pressure of the right foot, it is conceivable that a walking mode at a time of measuring the foot pressure of the left foot differs from a walking mode at a time of measuring the foot pressure of the right foot, for example, a walking speed and a step length differ. Although the foot pressures of the left foot and

the right foot are originally symmetrical, for example, the foot pressure of the left foot is substantially similar to the foot pressure of the right foot at the time of measuring the foot pressure of the left foot, there is a likelihood that the risk estimation apparatus **200** may erroneously determine that there is large asymmetry between the foot pressures in the left foot and the right foot because a measuring time of the foot pressure of the left foot differs from a measuring time of the foot pressure of the right foot and the walking modes differ.

(70) In contrast, it is expected that the risk estimation apparatus **200** can determine asymmetry between the foot pressures with higher accuracy by determining asymmetry using associated data of the foot pressures of the left foot and the right foot (data of the foot pressures synchronized by the synchronization unit **292**).

(71) The risk estimation unit **294** estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

(72) For example, the risk estimation unit **294** may estimate a lower limb abnormality risk on the basis of asymmetry between the sum of the foot pressures in the heels of the left foot and the right foot during the stance phase and the sum of the foot pressures in the toes of the left foot and the right foot during the stance phase. That is, the risk estimation unit **294** may estimate a lower limb abnormality risk on the basis of asymmetry between the sum of the foot pressure in the heel and the foot pressure in the toes of the left foot during the stance phase of the left foot and the sum of the foot pressure in the heel and the foot pressure in the toes of the right foot during the stance phase of the right foot.

(73) Moreover, for example, the risk estimation unit **294** may estimate a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phases of the left foot and the right foot in which parts of the measurement periods temporally overlap each other. That is, the risk estimation unit **294** may determine asymmetry between the foot pressures of the left foot and the right foot using the measured data of the foot pressure of the left foot during the stance phase of the left foot and the measured data of the foot pressure of the right foot during the stance phase of the right foot which are synchronized by the synchronization unit **292**.

(74) FIG. **6** is a diagram illustrating an example of measured values of foot pressures obtained by the sensor system **100**. The horizontal axis of FIG. **6** indicates times. The vertical axis of FIG. **6** indicates measured values of the foot pressures. The vertical axis of FIG. **6** indicates normalized values of foot pressures (normalized loads) with a reference value of the foot pressure as **100**.

(75) A line L**21** indicates an example of measured values of the pressure of the left foot obtained by the left side sensor system **101**. The line L**21** indicates the values obtained by summing the foot pressure in the toes of the left foot by the left toe side sensor **111** and the foot pressure in the heel of the left foot by the left heel side sensor **121**.

(76) A line L**22** indicates an example of measured values of the pressure of the right foot obtained by the right side sensor system **102**. The line L**22** indicates the values obtained by summing the foot pressure in the toes of the right foot by the right toe side sensor **112** and the foot pressure in the heel of the right foot by the right heel side sensor **122**.

(77) A threshold A is a threshold used for identifying the starting timing and the ending timing of the stance phase by the stance phase identification unit **293**. The stance phase identification unit **293** compares the foot pressure with the threshold A and determines a timing at which the foot pressure changes from a value smaller than the threshold A to a value larger than the threshold A as the starting timing of the stance phase. Moreover, the stance phase identification unit **293** determines a timing at which the foot pressure changes from a value larger than the threshold A to a value smaller than the threshold A as the ending timing of the stance phase.

(78) In the example of FIG. **6**, the foot pressure of the left foot indicated by the line L**21** changes from a value smaller than the threshold A to a value larger than the threshold A at a time THE. Accordingly, the stance phase identification unit **293** determines the time T**11L** as the starting

timing of the stance phase of the left foot.

(79) Moreover, the foot pressure of the left foot indicated by the line L21 changes from a value larger than the threshold A to a value smaller than the threshold A at a time T12L. Accordingly, the stance phase identification unit 293 determines the time T12L as the ending timing of the stance phase of the left foot.

(80) Furthermore, the foot pressure of the left foot indicated by the line L21 changes from a value smaller than the threshold A to a value larger than the threshold A at a time T21L. Accordingly, the stance phase identification unit 293 determines the time T21L as the starting timing of the stance phase of the left foot.

(81) A period from the time T11L at which the stance phase of the left foot starts to the time T21L at which the stance phase of the left foot starts again corresponds to one walking cycle. One walking cycle determined on the basis of the left foot is also referred to as one walking cycle of the left foot.

(82) Moreover, in the example of FIG. 6, the foot pressure of the right foot indicated by the line L22 changes from a value smaller than the threshold A to a value larger than the threshold A at a time T11R. Accordingly, the stance phase identification unit 293 determines the time T11R as the starting timing of the stance phase of the right foot.

(83) Moreover, the foot pressure of the right foot indicated by the line L22 changes from a value larger than the threshold A to a value smaller than the threshold A at a time T12R. Accordingly, the stance phase identification unit 293 determines the time T12R as the ending timing of the stance phase of the right foot.

(84) Furthermore, the foot pressure of the right foot indicated by the line L22 changes from a value smaller than the threshold A to a value larger than the threshold A at a time T21R. Accordingly, the stance phase identification unit 293 determines the time T21R as the starting timing of the stance phase of the right foot.

(85) A period from the time T11R at which the stance phase of the right foot starts to the time T21R at which the stance phase of the right foot starts again corresponds to one walking cycle. One walking cycle determined on the basis of the right foot is also referred to as one walking cycle of the right foot.

(86) In one walking cycle of the left foot from the time T11L to the time T21L and one walking cycle of the right foot from the time T11R to the time T21R, periods from the time T11R to the time T21L temporally overlap each other.

(87) For example, the stance phase identification unit 293 may extract data for each walking cycle of the left foot from the measured data of the foot pressure of the left foot and may extract data for each walking cycle of the right foot from the measured data of the foot pressure of the right foot.

(88) Moreover, the synchronization unit 292 may associate (synchronize) data for one walking cycle of the left foot with data for one walking cycle of the right foot starting during the one walking cycle of the left foot. For example, the synchronization unit 292 may associate the measured data of the foot pressure of the left foot during one walking cycle of the left foot from the time T11L to the time T21L with the measured data of the foot pressure of the right foot during one walking cycle of the right foot from the time T11R to the time T21R.

(89) In this case, the risk estimation unit 294 may estimate a lower limb abnormality risk the basis of asymmetry between the associated stance phases during one walking cycle.

(90) For example, one walking cycle of the left foot from the time T11L to the time T21L includes the stance phase from the time T11L to the time T12L. Moreover, one walking cycle of the right foot from the time T11R to the time T21R includes the stance phase from the time T11R to the time T12R. The risk estimation unit 294 may calculate the index value of asymmetry between the stance phase from the time T11L to the time T12L and the stance phase from the time T11R to the time T12R.

(91) Various values can be used as the index value of asymmetry calculated by the risk estimation

unit **294**. For example, the risk estimation unit **294** may calculate any of the following as the index value of asymmetry by comparing the foot pressure of the left foot indicated by the line **L21** and the foot pressure of the right foot indicated by the line **L22** to each other.

(92) (1) Difference Between Local Maximum Values of Foot Pressure Indicated by Point P.sub.1

(93) At a point P.sub.1, when the stance phase starts, the heel has contacted the ground but the toes have not yet contacted the ground or have not sufficiently contacted the ground, and the foot on the opposite side is in the swing phase (takes off the ground) so that the weight is concentrated on the heel of one foot and a large measured value of the foot pressure is obtained by the heel side sensor **120**.

(94) (2) Difference Between Local Maximum Values of Foot Pressure Indicated by Point P.sub.2

(95) At a point P.sub.2, when the stance phase ends, the heel floats, only the toes contacts the ground, and the foot on the opposite side has not yet touched the ground so that the weight is concentrated on the toes of one foot and a large measured value of the foot pressure is obtained by the toe side sensor **110**.

(96) (3) Difference Between Local Minimum Values of Foot Pressure Indicated by Point D

(97) At a point D, the foot sole of one foot (the shoe sole) entirely contacts the ground, and the weight is dispersed throughout the entire foot sole so that relatively small foot pressures are applied to the toe side sensor **110** and the heel side sensor **120**.

(98) (4) Difference Between Integral Values of Foot Pressure During One Walking Cycle

(99) In the example of FIG. 6, an integral value of the foot pressures during one walking cycle of the left foot is indicated as an area of a portion surrounded by the line **L21** and the horizontal axis during the period from the time **T11L** to the time **T21L**. Moreover, an integral value of the foot pressures during one walking cycle of the right foot is indicated as an area of a portion surrounded by the line **L22** and the horizontal axis during the period from the time **T11R** to the time **T21R**.

(100) It should be noted that in the swing phase, the foot pressure becomes zero or approximately zero. Hence, the risk estimation unit **294** may calculate the integral value of the foot pressures for only the stance phase.

(101) (5) Difference Between Average Values of Foot Pressures During Stance Phase

(102) For example, the risk estimation unit **294** may perform sampling of the foot pressures during the stance phase in a predetermined sampling cycle and may calculate the average value.

Alternatively, the risk estimation unit **294** may calculate the average value of the foot pressures by dividing the integral value of the foot pressures during the stance phase by a time.

(103) (6) Value Obtained by Applying Predetermined Arithmetic Operation Such as the Four Basic Operations of Arithmetic to Values Obtained from (1) to (5) Described Above

(104) The risk estimation unit **294** may calculate a “ratio (proportion)” between the foot pressure of the left foot and the foot pressure of the right foot from (1) to (6) described above, in place of the “difference” between the foot pressure of the left foot and the foot pressure of the right foot.

(105) In all the index values calculated from (1) to (5) described above, the larger the value, the greater asymmetry between the foot pressure of the left foot and the foot pressure of the right foot.

(106) The risk estimation unit **294** may determine one of predetermined classifications of risk levels that corresponds to the calculated index value.

(107) FIG. 7 is a diagram illustrating an example of risk level information stored in the storage unit **280**. The risk estimation unit **294** may estimate a lower limb abnormality risk using the risk level information.

(108) In the example of FIG. 7, the risk level information is configured as data in a tabular format, and one row corresponds to one level of a lower limb abnormality risk. Each row of the risk level information includes a field of “asymmetry” and a field of “lower limb abnormality level”.

(109) The field of “lower limb abnormality level” shows a level of a lower limb abnormality risk. In the example of FIG. 7, “level 1”, “level 2”, “level 3”, and so on are shown as the levels of a lower limb abnormality risk, and the larger the numerical value, the higher the risk.

(110) The field of “asymmetry” shows a classification of the index value calculated by the risk estimation unit **294** with a minimum value and a maximum value. For example, if an index value  $x$  calculated by the risk estimation unit **294** satisfies  $a \leq x < b$ , the risk estimation unit **294** determines that the lower limb abnormality risk is in “level 1”. If the index value  $x$  calculated by the risk estimation unit **294** satisfies  $b \leq x < c$ , the risk estimation unit **294** determines that the lower limb abnormality risk is in “level 2”. If the index value  $x$  calculated by the risk estimation unit **294** satisfies  $c \leq x < d$ , the risk estimation unit **294** determines that the lower limb abnormality risk is in “level 3”.

(111) In this manner, the risk estimation unit **294** estimates a lower limb abnormality risk, and thus the subject person can be informed of the lower limb abnormality risk. When the lower limb abnormality risk is high, the subject person can seek for countermeasures such as improving the gait to reduce asymmetry between the foot pressure of the left foot and the foot pressure of the right foot.

(112) It should be noted that the risk estimation unit **294** may estimate the level of a lower limb abnormality risk by a method other than referring to the risk level information, such as estimating the level of a lower limb abnormality risk through linear regression or machine learning.

(113) Moreover, the risk estimation unit **294** may estimate the level of a lower limb abnormality risk using data for a plurality of walking cycles synchronized between the left foot and the right foot. For example, the risk estimation unit **294** may estimate the level of a lower limb abnormality risk using the average value of the foot pressures for a plurality of walking cycles.

(114) Next, an operation of the risk estimation apparatus **200** will be described with reference to FIGS. **8** to **10**.

(115) FIG. **8** is a flowchart illustrating an example of a procedure of processing in which the risk estimation apparatus **200** estimates a lower limb abnormality risk.

(116) In the processing of FIG. **8**, the data acquisition unit **291** acquires the measured data of the foot pressure of the left foot and the measured data of the foot pressure of the right foot (step **S101**).

(117) Moreover, the synchronization unit **292** performs time adjustment between the measured data of the foot pressure of the left foot and the measured data of the foot pressure of the right foot (step **S102**). For example, when the time (time stamp) attached to the measured data of the foot pressure of the left foot is later than the time attached to the measured data of the foot pressure of the right foot, the synchronization unit **292** may perform time adjustment advancing the time attached to the measured data of the foot pressure of the left foot by the time difference between the time attached to the measured data of the foot pressure of the right foot and the time attached to the measured data of the foot pressure of the left foot.

(118) Next, the stance phase identification unit **293** identifies the starting timing of the stance phase and the ending timing of the stance phase for each of the measured data of the foot pressure of the left foot and the measured data of the foot pressure of the right foot (step **S103**). For example, as described with reference to FIG. **6**, the stance phase identification unit **293** may identify the starting timing of the stance phase and the ending timing of the stance phase by comparing the measured value of the foot pressure with the threshold.

(119) Through the processing of step **S103**, the measured data of the foot pressures is divided into pieces of data for each walking cycle. For example, the measured data of the foot pressures in the left foot during the period from the starting timing of the stance phase of the left foot to the starting timing of the next stance phase can be used as the measured data for one walking cycle of the left foot. The measured data of the foot pressures in the right foot during the period from the starting timing of the stance phase of the right foot to the starting timing of the next stance phase can be used as the measured data for one walking cycle of the right foot.

(120) Next, the synchronization unit **292** synchronizes the measured data for one walking cycle of the left foot with the measured data for one walking cycle of the right foot (step **S104**). For

example, as described with reference to FIG. 6, the synchronization unit **292** may synchronize (associate) the measured data for one walking cycle of the left foot with the measured data for one walking cycle of the right foot starting during the period of this walking cycle.

(121) Next, the risk estimation unit **294** calculates the index value of asymmetry between the foot pressures using the synchronized measured data of the foot pressures of the left foot and the right foot (the measured data for one walking cycle of the foot pressures in the left foot and the measured data for one walking cycle of the foot pressures in the right foot that is associated with the measured data for one walking cycle of the foot pressures in the left foot) (step **S105**). As described above, various index values can be used as the index value of asymmetry between the foot pressures.

(122) Further, the risk estimation unit **294** estimates a lower limb abnormality risk on the basis of the calculated index value (step **S106**). As described with reference to FIG. 7, the risk estimation unit **294** may estimate a lower limb abnormality risk using the risk level information. For example, the risk estimation unit **294** may cause the display unit **220** to display an estimated risk.

(123) After step **S106**, the risk estimation apparatus **200** ends the processing of FIG. 8.

(124) It should be noted that the processing procedure illustrated in FIG. 8 is an example of a procedure of processing performed by the risk estimation apparatus **200**, and a procedure of processing performed by the risk estimation apparatus **200** is not limited thereto. For example, when the risk estimation apparatus **200** receives the measured data of the foot pressures from each of the left side sensor system **101** and the right side sensor system **102** in real time and a time stamp is attached by the risk estimation apparatus **200** side, time adjustment (step **S102**) of the measured data is not necessary. In the processing described with reference to FIGS. 9 and 10, a procedure of processing performed by the risk estimation apparatus **200** is also not limited thereto.

(125) FIG. 9 is a flowchart illustrating an example of a procedure of processing in which the risk estimation apparatus **200** synchronizes data of the left side sensor system **101** with data of the right side sensor system **102**.

(126) In the processing of FIG. 9, the synchronization unit **292** synchronizes the time of the left side sensor system **101** with the time of the right side sensor system **102** (step **S201**). For example, when a timepiece for a time stamp is built into each of the left side sensor system **101** and the right side sensor system **102**, the synchronization unit **292** may adjust the time of the timepiece of the left side sensor system **101** and the time of the timepiece of the right side sensor system **102** to the time of the timepiece built into the synchronization unit **292** itself. Through the processing of step **S201**, the time attached to the measured data of the foot pressure of the left foot by the left side sensor system **101** is synchronized with the time attached to the measured data of the foot pressure of the right foot by the right side sensor system **102**.

(127) The processing of step **S201** corresponds to an example of the processing of step **S102** in FIG. 8.

(128) Next, the stance phase identification unit **293** starts monitoring of the measured data of the foot pressures in the left foot using the left side sensor system **101** (step **S202**).

(129) Further, the stance phase identification unit **293** determines the time at which the heel of the left foot contacts the ground and stores this time in the storage unit **280** as a starting time of the stance phase of the left foot (step **S203**). For example, as described with reference to FIG. 6, the stance phase identification unit **293** may detect the starting time of the stance phase (the time at which the heel contacts the ground) by comparing the measurement value of the foot pressure with the threshold.

(130) The time at which the heel of the left foot contacts the ground, which is determined in step **S203**, will be expressed as a time **T31L**.

(131) The stance phase identification unit **293**, which has detected the starting time of the stance phase of the left foot in step **S203**, ends monitoring of the measured data of the foot pressure of the left foot (step **S204**) and starts monitoring of the measured data of the foot pressure of the right foot

using the right side sensor system **102** (step **S205**).

(132) Then, the stance phase identification unit **293** determines the time at which the heel of the right foot contacts the ground and stores the time in the storage unit **280** as a starting time of the stance phase of the right foot (step **S206**).

(133) The time at which the heel of the right foot contacts the ground determined in step **S206** will be expressed as a time **T31R**.

(134) The stance phase identification unit **293**, which has detected the starting time of the stance phase of the right foot in step **S206**, ends monitoring of the measured data of the foot pressure of the right foot (step **S207**) and starts monitoring of the measured data of the foot pressure of the left foot again using the left side sensor system **101** (step **S208**).

(135) Then, the stance phase identification unit **293** determines the time at which the heel of the left foot contacts the ground and stores the time in the storage unit **280** as the starting time of the stance phase of the left foot next to the stance phase of the left foot detected in step **S203** (step **S209**).

(136) The time at which the heel of the left foot contacts the ground determined in step **S209** will be expressed as a time **T41L**.

(137) The stance phase identification unit **293**, which has detected the starting time of the stance phase of the left foot in step **S209**, ends monitoring of the measured data of the foot pressure of the left foot (step **S210**) and starts monitoring of the measured data of the foot pressure of the right foot again using the right side sensor system **102** (step **S211**).

(138) Then, the stance phase identification unit **293** determines the time at which the heel of the right foot contacts the ground and stores the time in the storage unit **280** as the starting time of the stance phase of the right foot next to the stance phase of the right foot detected in step **S206** (step **S212**).

(139) The time at which the heel of the right foot contacts the ground determined in step **S212** will be expressed as a time **T41R**.

(140) The stance phase identification unit **293**, which has detected the starting time of the stance phase of the right foot in step **S212**, ends monitoring of the measured data of the foot pressure of the right foot (step **S213**).

(141) The processing from step **S202** to step **S213** corresponds to an example of processing for identifying the starting timing of the stance phase in the processing of step **S103** in FIG. **8**.

(142) Next, the synchronization unit **292** attaches a synchronization flag to the data for one walking cycle of the left foot and the data for one walking cycle of the right foot (step **S214**). A synchronization flag is a flag indicating that pieces of data are associated with each other through synchronization.

(143) Specifically, the synchronization unit **292** associates the measured data of the foot pressures for one walking cycle of the left foot from the time **T31L** to the time **T41L** with the measured data of the foot pressures for one walking cycle of the right foot from the time **T31R** to the time **T41R** by applying a synchronization flag.

(144) The processing of step **S214** corresponds to an example of the processing of step **S104** in FIG. **8**.

(145) After step **S214**, the risk estimation apparatus **200** ends the processing of FIG. **9**.

(146) FIG. **10** is a flowchart illustrating an example of a procedure of processing in which the risk estimation apparatus **200** determines the stance phases for data of the left side sensor system **101** and data of the right side sensor system **102**. The risk estimation apparatus **200** performs the processing of FIG. **10** for each of the measured data of the foot pressure of the left foot and the measured data of the foot pressure of the right foot.

(147) The processing of FIG. **10** corresponds to an example of the processing of step **S103** in FIG. **8**.

(148) In the processing of FIG. **10**, the stance phase identification unit **293** refers to the measured data of the foot pressures in order of time and determines whether or not the foot pressure is larger

than the threshold (step S301).

(149) If the stance phase identification unit **293** determines that the foot pressure is equal to or smaller than the threshold (step S301: NO), the processing returns to step S301.

(150) In contrast, if the stance phase identification unit **293** determines that the foot pressure is larger than the threshold (step S301: YES), the stance phase identification unit **293** determines that it is a timing at which the heel contacts the ground and stores this time in the storage unit **280** (step S302).

(151) Next, the stance phase identification unit **293** further refers to the measured data of the foot pressures in order of time and determines whether or not the foot pressure is smaller than the threshold (step S303).

(152) If the stance phase identification unit **293** determines that the foot pressure is equal to or larger than the threshold (step S303: NO), the processing returns to step S303.

(153) In contrast, if the stance phase identification unit **293** determines that the foot pressure is smaller than the threshold (step S303: YES), the stance phase identification unit **293** determines that it is a timing at which the toes take off the ground and attaches a stance phase ending flag to the data at this time (step S304). Alternatively, the stance phase identification unit **293** may store this time in the storage unit **280** as the ending time of the stance phase.

(154) Next, the stance phase identification unit **293** further refers to the measured data of the foot pressures in order of time and determines whether or not the foot pressure is larger than the threshold (step S305).

(155) If the stance phase identification unit **293** determines that the foot pressure is equal to or smaller than the threshold (step S305: NO), the processing returns to step S305.

(156) In contrast, if the stance phase identification unit **293** determines that the foot pressure is larger than the threshold (step S305: YES), the stance phase identification unit **293** determines that it is a timing at which the heel contacts the ground next to the timing at which the heel contacts the ground detected in step S302 and stores this time in the storage unit **280** (step S306).

(157) After step S306, the risk estimation apparatus **200** ends the processing of FIG. 10.

(158) FIG. 11 is a diagram illustrating another example of a configuration of the risk estimation system 1.

(159) In the example of FIG. 11, the risk estimation apparatus **200** is configured to include a portable terminal apparatus **310** and a server apparatus **320**.

(160) For example, the portable terminal apparatus **310** is a device such as a smartphone which can communicate with each of the left side sensor system **101**, the right side sensor system **102**, and the server apparatus **320**.

(161) The server apparatus **320** is an apparatus that executes at least some of the functions of the risk estimation apparatus **200**. The server apparatus **320** is configured using a computer such as a personal computer or a workstation.

(162) In the configuration of FIG. 11, various kinds of allotment can be adopted as allotment of the functions of the risk estimation apparatus **200** between the portable terminal apparatus **310** and the server apparatus **320**.

(163) For example, the server apparatus **320** may execute the function of the risk estimation unit **294**, and the portable terminal apparatus **310** may execute the other functions of the risk estimation apparatus **200**. Alternatively, the portable terminal apparatus **310** may transfer data from the left side sensor system **101** and data from the right side sensor system **102** to the server apparatus **320** without any change, and the server apparatus **320** may execute each of the functions of the risk estimation apparatus **200**.

(164) As described above, the data acquisition unit **291** acquires the measured data of the foot pressures from each of the left side sensor system **101** and the right side sensor system **102**. The stance phase identification unit **293** identifies the starting timing and the ending timing of the stance phase of each of the left foot and the right foot from the measured data of the foot pressures.

The risk estimation unit **294** estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

(165) In this manner, the risk estimation apparatus **200** can estimate a risk related to the feet other than a relationship between the feet and the shoes **810**. The risk estimation apparatus **200** estimates a lower limb abnormality risk, and thus the subject person can be informed of the lower limb abnormality risk. When the lower limb abnormality risk is high, the subject person can seek for countermeasures such as improving the gait to reduce asymmetry between the foot pressure of the left foot and the foot pressure of the right foot.

(166) Moreover, in the risk estimation system **1**, sensors are provided in the shoes **810**, and thus there is no need for the subject person to separately wear a wristband or the like, for example. In this regard, with the risk estimation system **1**, it is possible to prevent the subject person from feeling inconvenience of attachment/detachment of the sensors or the like and the subject person from forgetting to wear the sensors.

(167) Moreover, the subject person does not feel that he/she is wearing sensors, and thus it is expected that the subject person walks with a gait as in walking at a normal time. In this regard, the risk estimation system **1** can estimate a lower limb abnormality risk with high accuracy.

(168) Moreover, the data acquisition unit **291** acquires the measured data of the foot pressures on the heel side and the toe side of each of the left foot and the right foot. The risk estimation unit **294** estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures on the heel side and the toe side of the left foot and the right foot during the stance phase.

(169) In this manner, the risk estimation unit **294** estimates a lower limb abnormality risk using both the foot pressures on the heel side and the foot pressures on the toe side, and thus even when asymmetry occurs between the foot pressures on only one of the heel side and the toe side, a lower limb abnormality risk can be estimated by reflecting the asymmetry. In this regard, the risk estimation apparatus **200** can estimate a lower limb abnormality risk with high accuracy.

(170) Moreover, the risk estimation unit **294** estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase of each of the left foot and the right foot in which parts of the measurement periods temporally overlap each other.

(171) Assuming a case in which a measurement time of the measured data of the foot pressure of the left foot significantly differs from a measurement time of the measured data of the foot pressure of the right foot, it is conceivable that a walking mode at a time of measuring the foot pressure of the left foot differs from a walking mode at a time of measuring the foot pressure of the right foot, for example, a walking speed and a step length differ. Although the foot pressures of the left foot and the right foot are originally symmetrical, for example, the foot pressure of the left foot is substantially similar to the foot pressure of the right foot at the time of measuring the foot pressure of the left foot, there is a likelihood that the risk estimation apparatus **200** may erroneously determine that there is large asymmetry between the foot pressures in the left foot and the right foot because a measuring time of the foot pressure of the left foot differs from a measuring time of the foot pressure of the right foot and the walking modes differ.

(172) In contrast, it is expected that the risk estimation unit **294** can determine asymmetry between the foot pressures with higher accuracy by determining asymmetry between the foot pressures using the foot pressures of the left foot and the right foot during the stance phase of each of the left foot and the right foot in which parts of the measurement periods temporally overlap each other (data of the foot pressures synchronized by the synchronization unit **292**).

(173) FIG. **12** is a diagram illustrating an example of a configuration of a risk estimation apparatus according to the example embodiment.

(174) In the configuration illustrated in FIG. **12**, a risk estimation apparatus **410** includes a data acquisition unit **411**, a stance phase identification unit **412**, and a risk estimation unit **413**.

(175) In the configuration, the data acquisition unit **411** acquires the measured data of the foot

pressures obtained by sensors which are provided in shoes and measure the foot pressures of the left foot and the right foot. The stance phase identification unit **412** identifies the starting timing and the ending timing of the stance phase of each of the left foot and the right foot from the measured data of the foot pressures. The risk estimation unit **413** estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

(176) In this manner, the risk estimation apparatus **410** can estimate a risk related to the feet other than a relationship between the feet and shoes. The risk estimation apparatus **410** estimates a lower limb abnormality risk and thus the subject person (the subject person for estimation of a lower limb abnormality risk) can be informed of the lower limb abnormality risk. When the lower limb abnormality risk is high, the subject person can seek for countermeasures such as improving the gait to reduce asymmetry between the foot pressure of the left foot and the foot pressure of the right foot.

(177) Moreover, the risk estimation apparatuses **410** uses sensors provided in shoes and thus there is no need for the subject person to separately wear a wristband or the like, for example. In this regard, with the risk estimation apparatus **410**, it is possible to prevent the subject person from feeling inconvenience of attachment/detachment of the sensors or the like and the subject person from forgetting to wear the sensors.

(178) Moreover, the subject person does not feel that he/she is wearing sensors, and thus it is expected that the subject person walks with a gait as in walking at a normal time. In this regard, the risk estimation apparatus **410** can estimate a lower limb abnormality risk with high accuracy.

(179) FIG. **13** is a diagram illustrating an example of a configuration of a risk estimation system according to the example embodiment.

(180) In the configuration illustrated in FIG. **13**, a risk estimation system **420** includes sensors **421**, a stance phase identification unit **422**, and a risk estimation unit **423**.

(181) In the configuration, the sensors **421** are provided in shoes and measure the foot pressures of the left foot and the right foot. The stance phase identification unit **422** identifies the starting timing and the ending timing of the stance phase of each of the left foot and the right foot from the measured data of the foot pressures. The risk estimation unit **423** estimates a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase.

(182) In this manner, the risk estimation system **420** can estimate a risk related to the feet other than a relationship between the feet and shoes. The risk estimation system **420** estimates a lower limb abnormality risk, and thus the subject person (the subject person for estimation of a lower limb abnormality risk) can be informed of the lower limb abnormality risk. When the lower limb abnormality risk is high, the subject person can seek for countermeasures such as improving the gait to reduce asymmetry between the foot pressure of the left foot and the foot pressure of the right foot.

(183) Moreover, the sensors **421** are provided in shoes, and thus there is no need for the subject person to separately wear a wristband or the like, for example. In this regard, with the risk estimation system **420**, it is possible to prevent the subject person from feeling inconvenience of attachment/detachment of the sensors or the like and the subject person from forgetting to wear the sensors.

(184) Moreover, the subject person does not feel that he/she is wearing sensors, and thus it is expected that the subject person walks with a gait as in walking at a normal time. In this regard, the risk estimation system **420** can estimate a lower limb abnormality risk with high accuracy.

(185) FIG. **14** is a flowchart illustrating an example of a procedure of processing in a risk estimation method of the risk estimation apparatus according to the example embodiment.

(186) In the processing of FIG. **14**, the risk estimation apparatus executes a step of acquiring measured data of foot pressures obtained by sensors that are provided in shoes and measure the foot

pressures of the left foot and the right foot (step **S401**), a step of identifying a starting timing and an ending timing of a stance phase of each of the left foot and the right foot from the measured data of the foot pressures (step **S402**), and a step of estimating a lower limb abnormality risk on the basis of asymmetry between the foot pressures of the left foot and the right foot during the stance phase (step **S403**).

(187) In this manner, with the processing of FIG. **14**, a risk related to the feet other than a relationship between the feet and shoes can be estimated. The risk estimation apparatus estimates a lower limb abnormality risk using the processing of FIG. **14** and thus the subject person (the subject person for estimation of a lower limb abnormality risk) can be informed of the lower limb abnormality risk. When the lower limb abnormality risk is high, the subject person can seek for countermeasures such as improving the gait to reduce asymmetry between the foot pressures in the left foot and the foot pressure of the right foot.

(188) Moreover, in the processing of FIG. **14**, a risk estimation apparatus uses sensors provided in shoes, and thus there is no need for the subject person to separately wear a wristband or the like. In this regard, with the processing of FIG. **14**, it is possible to prevent the subject person from feeling inconvenience of attachment/detachment of the sensors or the like and the subject person from forgetting to wear the sensors.

(189) Moreover, the subject person does not feel that he/she is wearing sensors, and thus it is expected that the subject person walks with a gait as in walking at a normal time. In this regard, in FIG. **14**, a lower limb abnormality risk can be estimated with high accuracy.

(190) FIG. **15** is a schematic block diagram illustrating a configuration of a computer according to at least one of the example embodiments.

(191) In the configuration illustrated in FIG. **15**, a computer **700** includes a CPU **710**, a main storage apparatus **720**, an auxiliary storage apparatus **730**, and an interface **740**.

(192) At least one of the risk estimation apparatus **200** and the risk estimation apparatus **410** may be implemented in the computer **700**. In such a case, operation of each of the processing units described above is stored in the auxiliary storage apparatus **730** in a form of a program. The CPU **710** reads the program from the auxiliary storage apparatus **730**, loads the program in the main storage apparatus **720**, and executes the foregoing processing in accordance with the program. Moreover, the CPU **710** reserves storage areas corresponding to the storage units described above in the main storage apparatus **720** in accordance with the program. Communication between each of the apparatuses and other apparatuses is executed by having the interface **740** with a communication function and performing communication in accordance with control of the CPU **710**.

(193) When the risk estimation apparatus **200** is implemented in the computer **700**, operation of the control unit **290** and each unit thereof are stored in the auxiliary storage apparatus **730** in a form of a program. The CPU **710** reads the program from the auxiliary storage apparatus **730**, loads the program in the main storage apparatus **720**, and executes the foregoing processing in accordance with the program.

(194) Moreover, the CPU **710** reserves a storage area corresponding to the storage unit **280** in the main storage apparatus **720** in accordance with the program. Communication performed by the communication unit **210** is executed by having the interface **740** with a communication function and performing communication in accordance with control of the CPU **710**. The function of the display unit **220** is executed by having the interface **740** with a display screen and displaying an image in the display screen in accordance with control of the CPU **710**. The function of the operation input unit **230** is executed by having the interface **740** with an input device and receiving an operation of a user.

(195) When the risk estimation apparatus **410** is implemented in the computer **700**, operation of the data acquisition unit **411**, the stance phase identification unit **412**, and the risk estimation unit **413** is stored in the auxiliary storage apparatus **730** in a form of a program. The CPU **710** reads the

program from the auxiliary storage apparatus **730**, loads the program in the main storage apparatus **720**, and executes the foregoing processing in accordance with the program.

(196) It should be noted that the processing of each unit may be performed by recording a program for executing all or part of the processing performed by the risk estimation apparatus **200** and the risk estimation apparatus **410** in a computer readable recording medium and causing a computer system to read and execute the program recorded in this recording medium. It should be noted that a “computer system” here includes an operating system (OS) and hardware such as peripheral devices.

(197) Moreover, a “computer readable recording medium” refers to a portable medium such as a flexible disk, a magneto-optical disc, a read only memory (ROM), or a compact disc read only memory (CD-ROM); or a storage apparatus such as a hard disk built into a computer system. Moreover, the foregoing program may be a program for realizing some of the functions described above, and may be a program capable of realizing the functions described above in combination with a program which has already been recorded in a computer system.

(198) While the present invention has been particularly shown and described with reference to example embodiments thereof, the present invention is not limited to these example embodiments. It will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present invention as defined by the claims.

## Claims

1. A risk estimation apparatus comprising: sensors measuring foot pressures of a left foot and a right foot of a user, the sensors respectively provided in a left insole of a left shoe and a right insole of a right shoes of the user; at least one memory storing instructions; and at least one processor configured to execute the instructions to: acquire, from the sensors, measured data indicating a time variation of the foot pressures of the left foot and the right foot, wherein the measured data of the foot pressures of the left foot and the right foot are acquired on both a heel side and a toe side of each of the left foot and the right foot; synchronize the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot; calculate an index value of asymmetry between the foot pressures on the heel side and the toe side of the left foot and the right foot based on a local maximum value and a local minimum value of the synchronized measured data of the foot pressures of the left foot and the right foot; estimate a risk of abnormality of a lower limb by using a machine learned model on the basis of the index value of the asymmetry between the foot pressures of the left foot and the right foot, wherein the risk of the abnormality of the lower limb is used for making a decision related to improving a gait of the user to reduce asymmetry between the foot pressures of the left foot and the right foot.
2. The risk estimation apparatus according to claim 1, wherein the at least one processor is further configured to execute the instructions to synchronize the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot by associating data for one walking cycle of the left foot with data for one walking cycle of the right foot with respect to parts of measurement periods that overlap each other.
3. The risk estimation apparatus according to claim 1, wherein two sensors are provided in the left shoe and two sensors are provided in the right shoe.
4. The risk estimation apparatus according to claim 3, wherein one sensor of the two sensors provided in the left shoe is installed on a toe side of the left shoe, and another sensor of the two sensors provided in the left shoe is installed on a heel side of the left shoe, and wherein one sensor of the two sensors provided in the right shoe is installed on a toe side of the right shoe, and another sensor of the two sensors provided in the right shoe is installed on a heel side of the right shoe.
5. The risk estimation apparatus according to claim 1, wherein the sensors provided in the left shoe

and the right shoe are installed in a bilaterally symmetrical manner with respect to each other.

6. A risk estimation method executed by a computer and comprising: acquiring, from sensors measuring foot pressures of a left foot and a right foot and respectively provided in a left insole of a left shoe and a right insole of a right shoe of a user, measured data indicating a time variation of the foot pressures of the left foot and the right foot, wherein the measured data of the foot pressures of the left foot and the right foot are acquired on both a heel side and a toe side of each of the left foot and the right foot; synchronizing the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot; calculating an index value of asymmetry between the foot pressures on the heel side and the toe side of the left foot and the right foot based on a local maximum value and a local minimum value of the synchronized measured data of the foot pressures of the left foot and the right foot; estimating a risk of abnormality of a lower limb by using a machine learned model on the basis of the index value of the asymmetry between the foot pressures of the left foot and the right foot, wherein the risk of the abnormality of the lower limb is used for making a decision related to improving a gait of the user to reduce asymmetry between the foot pressures of the left foot and the right foot.

7. The risk estimation method according to claim 6, further comprising synchronizing the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot by associating data for one walking cycle of the left foot with data for one walking cycle of the right foot with respect to parts of measurement periods that overlap each other.

8. The risk estimation method according to claim 6, wherein two sensors are provided in the left shoe and two sensors are provided in the right shoe.

9. The risk estimation method according to claim 8, wherein one sensor of the two sensors provided in the left shoe is installed on a toe side of the left shoe, and another sensor of the two sensors provided in the left shoe is installed on a heel side of the left shoe, and wherein one sensor of the two sensors provided in the right shoe is installed on a toe side of the right shoe, and another sensor of the two sensors provided in the right shoe is installed on a heel side of the right shoe.

10. The risk estimation method according to claim 6, wherein the sensors provided in the left shoe and the right shoe are installed in a bilaterally symmetrical manner with respect to each other.

11. A non-transitory storage medium storing a program executable by a computer to perform a risk estimation method comprising: acquiring, from sensors measuring foot pressures of a left foot and a right foot and respectively provided in a left insole of a left shoe and a right insole of a right shoe of a user, measured data indicating a time variation of the foot pressures of the left foot and the right foot, wherein the measured data of the foot pressures of the left foot and the right foot are acquired on both a heel side and a toe side of each of the left foot and the right foot-obtained; synchronizing the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot; calculating an index value of asymmetry between the foot pressures on the heel side and the toe side of the left foot and the right foot based on a local maximum value and a local minimum value of the synchronized measured data of the foot pressures of the left foot and the right foot; estimating a risk of abnormality of a lower limb by using a machine learned model on the basis of the index value of the asymmetry between the foot pressures of the left foot and the right foot, wherein the risk of the abnormality of the lower limb is used for making a decision related to improving a gait of the user to reduce asymmetry between the foot pressures of the left foot and the right foot.

12. The risk estimation method according to claim 11, wherein the method further comprises synchronizing the measured data of the foot pressure of the left foot with the measured data of the foot pressure of the right foot by associating data for one walking cycle of the left foot with data for one walking cycle of the right foot with respect to parts of measurement periods that overlap each other.

13. The non-transitory storage medium according to claim 11, wherein two sensors are provided in the left shoe and two sensors are provided in the right shoe.

14. The non-transitory storage medium according to claim 13, wherein one sensor of the two sensors provided in the left shoe is installed on a toe side of the left shoe, and another sensor of the two sensors provided in the left shoe is installed on a heel side of the left shoe, and wherein one sensor of the two sensors provided in the right shoe is installed on a toe side of the right shoe, and another sensor of the two sensors provided in the right shoe is installed on a heel side of the right shoe.

15. The non-transitory storage medium according to claim 11, wherein the sensors provided in the left shoe and the right shoe are installed in a bilaterally symmetrical manner with respect to each other.

---